PAANIPULSE (पानीय पल्स)

India's Community-Powered Water Intelligence & Early Disruption Detection Network
Competition: Techfest 2026–27 Campus to Cloud Hackathon sponsored by AWS | **Round 1 Idea Brief**

Tagline: *"Know your neighbourhood's water situation before the tap runs dry."*
Live AWS Prototype URL: https://main.d16h9tm50mzqg3.amplifyapp.com
GitHub Repository: https://github.com/adiTya2106-thakare/paanipulse-water-intelligence
Core Technology: 100% Serverless AWS Architecture (Amazon API Gateway, AWS Lambda Python 3.12, Amazon DynamoDB, AWS Amplify)

1. Problem Statement & Indian Urban Context

In Indian urban and peri-urban habitats, municipal water supply is intermittent, operating for only 45 minutes to 3 hours a day with fluctuating pressure. Over 350 million urban residents across cities like Mumbai, Pune, Bengaluru, and New Delhi rely on shared rooftop tanks and underground sumps. When taps fail to flow, residents face immediate anxiety: Is it an internal motor failure or a ward-wide pipe burst? Today, residents rely on scattered WhatsApp messages and unverified rumors.

2. Proposed Solution

PAANIPULSE transforms informal neighbourhood chatter into a structured, transparent civic water-intelligence platform. In under 15 seconds, residents log tap availability and pressure. PAANIPULSE aggregates these observations, compares them against a diurnal **Neighbourhood Water Fingerprint**, computes a transparent **Community Signal Confidence** rating, and detects **Early Disruption Anomalies**.

3. Unique Differentiating Features

- **Neighbourhood Water Fingerprint:** Historical baseline by diurnal windows (Morning 6–10 AM, Midday 10 AM–4 PM, Evening 4–9 PM, Night 9 PM–6 AM). Cleanly flags when a baseline is still developing.
- **Transparent Community Signal Confidence:** Objective 100-point model based on reporter diversity (30 pts), recency decay (35 pts), and concordance agreement (35 pts).
- **Explainable Disruption Early-Warning:** Alerts residents when recent flow drops ≥ 50% below baseline with plain-language explanations. Never fabricates "official outage" claims.
- **Locality Pulse Map:** Privacy-preserving spatial explorer using municipal ward centroids (zero private home addresses stored).
- **Evidence-Based Locality Report Card:** Instant downloadable/printable audit card for RWAs and ward councillors.
- **Anti-Manipulation Integrity Layer:** Anonymous device token rate-limiting (max 3 reports / 15 mins), server-side timestamp validation, and XSS sanitization.

4. Cloud Architecture & Cost Assessment

PAANIPULSE operates on a 100% serverless AWS architecture: Amazon API Gateway routes requests, AWS Lambda executes Python 3.12 models, Amazon DynamoDB stores records using composite partition and sort keys to avoid table scans, and AWS Amplify delivers the frontend globally. The entire architecture operates at **$0.00/month** under the AWS Free Tier.

5. Expected Social Impact (Bharat Impact: 30%)

Prevents panic and private tanker price gouging, protects residential motors from burning out when running dry, and provides verifiable evidence report cards for citizens to present to local water boards.

PAANIPULSE — AWS Build Summary

AWS Cloud Architecture, Database Key Design, Serverless Implementation & Free Tier Cost Assessment
Techfest 2026–27 Campus to Cloud Hackathon sponsored by AWS

1. AWS Cloud Services Architecture

Service	Implementation State	Role in PAANIPULSE
Amazon API Gateway (HTTP API)	Implemented & Tested	CORS preflight handling, route dispatching, HTTPS termination, and burst rate-limiting.
AWS Lambda	Implemented & Tested	Python 3.12 execution engine running statistical confidence calculations, anomaly detection, and data validation.
Amazon DynamoDB	Implemented & Tested	On-demand NoSQL table `PaanipulseReports` with composite keys: PK `localityId`, SK `timestamp#reportId`.
Amazon CloudWatch	Implemented & Tested	Centralized serverless logging with 14-day retention.
AWS Amplify Hosting	Configured	Edge-accelerated distribution of compiled React 19 / TypeScript SPA.

2. DynamoDB Schema & Query Optimization

To eliminate expensive table scans, the table uses a composite key pattern:

- **Partition Key (HASH):** `localityId` (String)
- **Sort Key (RANGE):** `timestamp#reportId` (String)

All locality queries execute directly via indexed `Query` calls on the partition key, yielding consistent 10–15 ms response times.

3. Free Tier Cost Verification

The entire production stack operates at **$0.00 / month** under standard AWS Free Tier allowances (25 GB free DynamoDB, 1M free Lambda invocations, 1M free API Gateway calls).

4. Automated Test Verification

100% Pass Rate: 16 Unit & Algorithm Tests + 4 End-to-End Integration Tests executed with zero failures.

PAANIPULSE — Technical Architecture & Cloud Specification

100% Serverless Cloud Infrastructure on AWS • Techfest 2026–27 Campus to Cloud Hackathon

1. Architectural Principles

- **Zero-Idle-Cost Serverless:** Amazon API Gateway, AWS Lambda (Python 3.12), and Amazon DynamoDB.
- **Deterministic Civic Intelligence:** 100-point transparent Community Signal Confidence scoring.
- **Explainable Anomaly Detection:** Diurnal Neighbourhood Water Fingerprint comparison without black-box AI.
- **Privacy-by-Design:** Municipal ward and locality centroids only; zero private home addresses stored.

2. DynamoDB Key Optimization

Partition Key: `localityId` (S) | Sort Key: `timestamp#reportId` (S). Zero table scans; single-digit millisecond query execution.

3. Free Tier Cost Verification

Estimated monthly infrastructure cost: **$0.00 / month** under standard AWS Free Tier.